



decreases complexity on production lines and accelerates the market entry.

Sonic Edge

<https://sonicedge.io>

The world's first camera-lidar fusion sensor

Kyocera Corporation has launched the camera-lidar fusion sensor, which the company claims is the first lidar to integrate the optical axes of the camera and lidar into a single unit. The design allows the real time acquisition



of parallax-free superimposed data, a previously impossible capability. It also has the highest laser irradiation density among lidar sensors, improving its ability to detect objects at long distances precisely. The camera-lidar fusion sensor merges camera images and lidar distance data in real time, enhancing object recognition capabilities. Its resolution is enhanced by increasing the density of laser beams emitted, which improves both resolution and accuracy. With an irradiation density of 0.045 degrees, the sensor can detect small obstacles, such as a 30cm object at a distance of 100 metres.

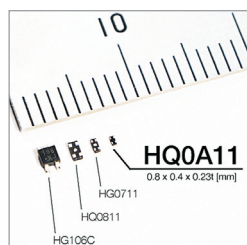
Kyocera Corporation

<https://global.kyocera.com>

Smallest and thinnest Hall element

Asahi Kasei Microdevices (AKM) has launched an InAs (indium arsenide) compound Hall element, the HQ0A11. With dimensions of 0.8mm x 0.4mm x 0.23mm, it shows an 85 % reduction in volume compared to its predecessor. The HQ0A11 is expected to improve

functionality in smartphone camera modules and compact robots, among other technologies. It offers an improved



S/N (signal-to-noise) ratio, approximately 16 % better than the HQ0811, which leads to better position detection accuracy. It has applications ranging from automotive airbag sensors to smartphone camera stabilisation and autofocus. It excels in S/N performance, with over 3.5 times the accuracy of AKM's previous GaAs Hall elements. This makes the HQ0A11 effective in reducing lens shake, which is particularly important in smartphone telephoto lenses.

Asahi Kasei Microdevices (AKM)

<https://www.asahi-kasei.com>

World's first commercial-grade radio processing unit

Digital Enhancement has launched the world's first commercial-grade radio processing unit (RPU) for wireless Wi-Fi access. The RPU features advanced



closed-loop training, real time calibration architecture, and a high-performance hybrid analogue-

to-digital-to-analogue pre-distortion algorithm, dramatically improving RF system performance using sophisticated digital compensation technology. In practical terms, Wi-Fi routers with integrated RPU provide extended high-speed coverage, while bridge devices benefit from higher speeds and longer ranges. The RPU also increases power amplifier output efficiency, reducing the need for costly and bulky directional antennas by lowering antenna gain requirements. It offers reliable anti-ageing performance for industrial Wi-Fi applications and maintains stable power output.

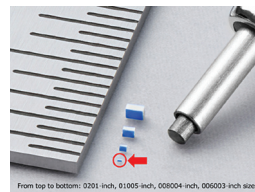
Digital Enhancement

www.digienh.com

World's smallest chip inductor

Murata Manufacturing Co. Ltd has launched a 006003-inch chip inductor,

the smallest in its class. This inductor achieves a volume reduction of about 75 % compared to the previously



available 008004-inch product. Due to the need for smaller components in electronic

mobile devices, engineers face challenges fitting more components into limited board space. Chip inductors, widely used in electronic devices, are increasingly incorporated in hundreds of smartphones. The chip inductor supports the miniaturisation and functionality of future electronic devices.

Murata

<https://www.murata.com>

Wireless audio SoC incorporating multiple functions

Airoha Technology has launched the AB1595, a wireless audio chip integrating multiple functions into a single system-on-chip (SoC). The SoC enhances speech noise reduction using AI algorithms and can process signals from up to 10 microphones, improving the range and quality of environmental



noise cancellation. This allows for real-time adjustments to the noise cancellation filters, ensuring a consistent experience

across different wearing styles. The chip features adaptive active noise cancellation (ANC) that dynamically adjusts to the user's fit and the surrounding noise, optimising comfort and noise reduction for a better user experience. The company claims it is the first in the industry to use a 6-core architecture for Bluetooth and AI audio processing, effectively reducing power consumption. The chip supports up to 12 hours of usage with an activated hearing aid function.

AIROHA

<https://www.airoha.com>